From eye movements to fixation-related potentials

How fixation onsets become trial-level EEG responses and pattern-level representations

1 Record EEG and gaze

EEG: 64 electrodes, 2,048 Hz
EyeLink: dominant eye, 2,000 Hz
Match fixation times to trial EEG

2 Preprocess continuous EEG

Average reference → notches at 50, 100, 150, 200, 250 Hz → ICA

Band-pass 0.1–100 Hz → interpolate bad electrodes → average reference

ICA fitted on a 1–100 Hz copy; retain components labeled “brain”.

Clean continuous EEG

3 Select sequence fixations

During the decision phase





Retain

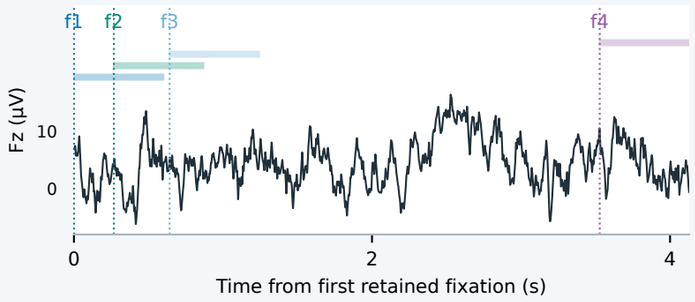
Mean gaze inside a sequence box
Fixation duration ≥ 100 ms

Exclude response-option fixations

Letter labels simplify the actual trial.

4 Extract fixation-locked EEG

One 0–600 ms epoch per retained fixation



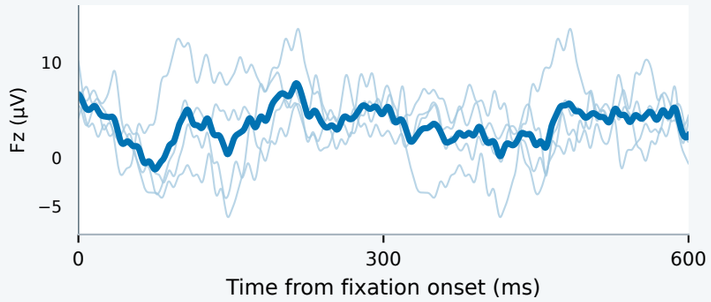
Start at fixation onset (t = 0)

Fixed window, even if fixation ends sooner
No fixation-specific baseline correction

Windows can overlap in continuous time.

5 Average within each trial

Align all retained epochs at fixation onset



One trial-level FRP per electrode

Each retained fixation contributes equally
Trials with no valid epochs have no FRP

Pale: individual epochs • Blue: their mean

6 Turn trial FRPs into representational geometry

Repeat separately for frontal, temporal, central, occipital and parietal channel groups.

REGIONAL TIME COURSE

Average electrodes
at each time point

PATTERN TIME COURSES

Average trials within
pattern and session

SESSION RDM

8 × 8 distances
Correlation distance: 1 − r

PARTICIPANT RDM

Average that participant’s
session RDMs

GROUP RDM

Average participant RDMs
with equal weights

FRPs summarize ongoing sequence inspection; they are not isolated responses to a single visual event.

Panels 4–5: measured EEG, participant 1 / session 1 / trial 1, electrode Fz; all four retained sequence fixations. Panel 3 uses simplified symbols for the same trial.